AGRICULTURE 4.0: Agriculture And Environment Monitoring

SANI THEO

Sensors empower farmers to react quickly and dynamically maximise crop performance

Global population growth, global warming, climate change and food security are among the most challenging problems around the world. Farmers need to increase their yields by about 50 per cent by 2050 to be able to feed the world, says Food and Agriculture Organization (FAO) of the United Nations (UN).

As population grows, the amount of cultivable land shrinks. To increase agricultural output from available arable land, one option is to invest in technology to meet the global demand for food. Various stages of the agricultural revolution are aimed at developing new ideas around sustainability, food production, energy and agriculture technologies.

Sensors and digital imaging technologies give farmers a better picture of their fields and crops. Data collected using these prove useful in improving crop yields and efficiency. Development of connectivity with various agricultural tools and sensor technologies lead to important progress in agricultural practice.

Sensors for agriculture include optical devices, sensors for crop health status determination, seed monitoring, detection of microorganisms and pest management, yield estimation and prediction, detection of crops, weeds and fruits, and airborne, soil, wearable, weather and Internet of Things (IoT) sensors, electronic noses and tongues, sensor networks and so on. This article covers some modern sensor systems used in the agriculture industry, smart farming and environment monitoring systems.

Agriculture 4.0

Agriculture has, for many years, been ignored and remained without modern technology. Now, it is making a big comeback with the use of different technologies. With connectivity gaining ground, we have GPS precision-guided tractors, sensor-based water irrigation systems, pest surveillance from air, smart livestock monitoring along with farmers hooked up to Big Data. We are witnessing agricultural innovation and adapting to a changing world and new settings including digital precision and smart farming.

Agriculture 4.0 is a new approach towards farm management and precision agriculture using technology, including sensors, smart tools, satellites, the IoT, remote sensing and proximal data gathering. It is about connectivity, smart agriculture or digital farming, in addition to the integrated internal and external networking of farm operations.

Agriculture 4.0 is said to have started around early 2010s, with low-cost and improved sensors, actuators, microprocessors, nanotechnology, high-bandwidth cellular communication, cloud computing and artificial intelligence (AI).

Fig. 1: Cognitive smart farming (Credit: www.ibm.com)

Fig. 2: Digital farming (Credit: www.ey.com)